BRAC University
Department of Computer Science and Engineering

Examination: Quiz 01 | Semester : Spring26
Duration: 20 Minutes | Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:12

1.

Source Code:

```
1. #include <stdio.h>
2. int g = 1;
3. int arr[4] = {2, 0, 2, 5};
4. int step(int a) {
5.     int j = g + arr[a];
6.     int m = 0;
7.     while (j < a) {
8.         m = 4 + a;
9.         a = a - 2;
10.       }
11.       return m;
12.   }
13.   void main(void) {
14.      float r = step(3);
15.      r = r + j;
16.      printf("%d\n", r);
17.      return 0;
18.   }
```

a. In a C-like compiler pipeline, why does the lexical analyzer 'discard' (or never tokenize) a line such as **line no. 1 of the given source code**? ***(5)***

b. How can the compiler still 'bring it back' so that diagnostics and tools know that an include has occurred? **(5)**

2. Explain with example why an ambiguous context free grammar can not be used in Syntax Analyzer ***(5)***

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